Exp No: 05

Capturing And Analyzing Packets Using Wireshark

Aim:

To capture, filter, inspect and analyze network packets using the wireshark network protocol analyzer.

Wireshark:

Wireshark is an open-source network protocol analyzer that captures packets in real time and displays them in a human-readable format.

It helps network administrators, developers and analyze different protocols using filters and packets.

Algorithm:

* Install and launch Wireshark.
* Select the active network interface (Ethernet).
* Set Capture → Stop capture automatically after 100 packets.
* Click Start capture.
* Capture the required packets.
* Apply display filters (TCP/UDP, ARP, DNS, HTTP)
* Inspect packet details and bytes.
* View the Flow Graph.
* Save the captured packets.

Features Of Wireshark:

* Captures live network traffic.
* Supports protocol analysis.
* Provide capture and display filters.

Main Components:

1) Packet List Pane: Displays all captured packets with information such as source, destination, protocol, length and time.

2) Packet Details Pane: Displays detailed protocols headers and fields of the selected packet.

3) Packet Bytes Pane: Displays the raw packet data in hexadecimal and ASCII format.

FILTERS:

* tcp || udp

Procedure:

- Start packet capture
- Apply the filter
- Inspect TCP/UDP packets
- View Statistics → Flow Graph
- Save the capture.

* arp

Procedure:

- Start packet capture.
- Apply the ARP filter.
- Inspect packets
- Save the capture.

* dns.

* http * dhcp * ip || icmp

RESULT:

Thus, the capturing, analyzing and filtering packets using Wireshark is successfull.

